

PYRANTIS

Which country burns in the Northern Territory, and when

Unit

PRT565 Machine Learning, Artificial Intelligence and Algorithms

Unit code

PRT565

Assessment

Assessment 3 – Group Presentation (30% of the unit)

Group number

Group 85

Campus

Darwin (Danala) Campus

Due

27 September 2026, 22:00 Sydney time

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The recorded presentation



Link to the recorded presentation

https://youtu.be/c_0yCsdZX2U

The link is set so that anyone with it can view the file. It was tested from a signed-out browser before submission.

Project title

PYRANTIS — multi-class classification of Northern Territory fire regimes, and a forecast for 2026.

What the project does

For every five-kilometre square of the Northern Territory, for every year from 2000 to 2025, the project classifies what happened into one of three answers: no fire, a fire first detected on or before 31 July, or a fire first detected after it. Six classifiers are trained on the same 966,819 rows and compared on the same held-out years, and the strongest of them is then used to publish a forecast for 2026 that can be checked against the satellite record in 2027.

The models are fed 50 features built from three public sources: satellite fire scars from NAFI, daily weather from the Queensland Government's SILO archive, and MODIS vegetation greenness obtained through Google Earth Engine. Every feature is restricted to information that existed by 30 April of the year being predicted, so a forecast arrives before the fire season rather than describing it afterwards.

Why we chose this topic

Fire in the Northern Territory is not an emergency that arrives occasionally. Across 2000 to 2025 about a third of the Territory carried a fire in any given year, so the useful question is not whether country will burn but when. Before August the country is still damp: fires are cool, they creep, and rangers light them deliberately to break up the fuel. After July the grass has cured, and the same ignition runs with the wind, burns hot enough to kill canopy, and takes out fences, stock and country in an afternoon.

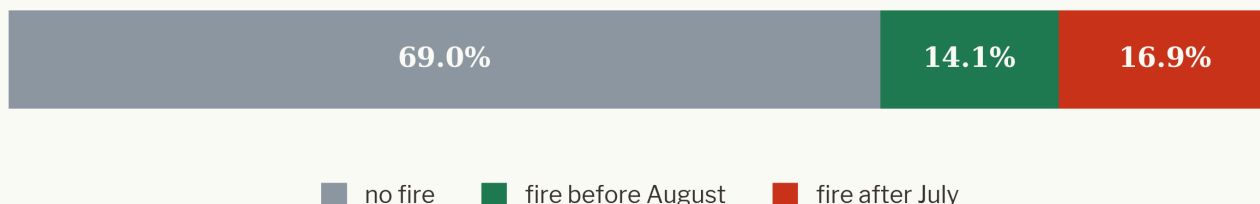
That difference is a decision someone makes in April, which is what made the problem worth modelling. A classifier that says which squares are likely to carry a late fire, while there is still time to plan a burning programme, answers a question a land manager already asks. The topic also sits where the unit is taught, and all three data sources are open, so every number in this report can be reproduced from the scripts in the repository.

Dataset description

Three archives are joined onto one grid of 0.05-degree cells, roughly five kilometres across. The grid is the SILO weather grid, so weather joins to fire without resampling either one.

■ NAFI	North Australian Fire Information, Charles Darwin University. 250-metre burnt-area rasters for 2000 to 2025. Each pixel carries the month the fire was detected, which is what makes an early or late label possible at all (North Australian Fire Information, 2026).
■ SILO	Queensland Government Data Drill. Daily rainfall, maximum temperature, vapour pressure deficit and solar radiation, interpolated from Bureau of Meteorology station observations onto the same grid (Jeffrey et al., 2001; Queensland Government, 2026).
■ MODIS	MOD13Q1 vegetation index at 250 m on a 16-day cycle, extracted through Google Earth Engine (Didan, 2021; Gorelick et al., 2017). Greenness over the wet season measures how much fuel is standing when the dry arrives.

Labelling produced 1,207,570 cell-years across 46,445 cells. Cells on the coast and the state border have no complete neighbourhood, and the first five years carry too little history for the longest look-back windows, so the modelling table holds 966,819 rows over 2005 to 2025.



Class balance across the modelling rows. The answer that matters most, a fire after July, is also close to the rarest.

Roughly seven rows in ten record no fire. An accuracy figure alone would therefore reward a model that never predicts a fire at all, which is why the comparison in Section 4 is led by macro-averaged F1 and why the models are trained with class weights.

Analysis, preprocessing and feature engineering

From burnt pixels to one answer per square

Each NAIFI pixel holds the month it burned, from 1 to 12, or zero for no fire. Every 20 by 20 block of pixels is pooled into one cell, and the fraction that burned before August and the fraction that burned after July are counted separately. A cell counts as burnt only when at least 5% of it burned, and it enters the table at all only when at least half its pixels were readable through cloud. Whichever season holds more of the burnt area names the year.

No metric can catch a wrong label, so the labels were checked against fire history that is already published. The pipeline puts 2011, the year after the 2010–11 La Niña, at 65.7% of the map burnt, the heaviest year on record, and 2020 at 15.9%, the lightest. Both agree with the published record (Russell-Smith et al., 2007).

What the models are allowed to know

Every one of the 50 features is built from information available by 30 April of the year being predicted. They fall into four groups: 26 fire-history features, 14 weather features, 7 greenness features and 3 that place the square.

■ Fire history (26)	Years since the last fire and the last late fire, how often the square burnt over 3, 5, 10, 15, 20 and 25 years, what it did in each of the two previous years, and what its eight neighbours did last year.
■ Weather (14)	Wet-season rainfall, heat, vapour pressure deficit and radiation, the dry season before it, days since rain, and how far the wet sat from that cell's own local normal.
■ Greenness (7)	Mean, maximum and minimum greenness over the wet, the April reading, how fast the country greened up, and two anomalies against the cell's own history.
■ Place (3)	Longitude, latitude and elevation, so the model can separate the Top End from the arid south without being handed the regions.

Why these features

The starting list came from how savanna fires work. Fuel builds up in the years after a fire, and a wetter wet season grows more grass. Fire history therefore tells the model how much fuel has built up since the last fire, because country that burned last year has little grass left to carry another. Weather and greenness tell it how much grass the wet season grew and how dry that grass is likely to get. Location lets it tell the wet Top End from the arid south.

Two kinds of input were left out on purpose. Anything measured in the year being predicted, such as that year's burnt area, would hand the model the answer. Each square's long-term average greenness was also dropped, because it works like a name tag for the square and says nothing about the year being predicted.

Each idea was kept only if it improved the score on years the model had not seen. Section 3 describes the 23 experiments that decided which extra features to keep, and Section 4 shows what each group adds to the score and which inputs the models rely on most. About 14 of the 50 features barely change the score, such as days since the last rain, April rainfall and whether a square has ever burned. Some of them overlap with other features, so the model gets the same information elsewhere. The list works well in testing but has not been shown to be the best possible set.

Two tests against leakage

A feature that quietly reads the year it is meant to predict scores well and means nothing, so the pipeline was tested for it twice rather than trusted. First, every fire-history feature is rebuilt from years strictly before the target year inside a single function that both the training path and the forecast path call; written twice, the two copies would drift apart, and a forecast has no label to catch the drift. Second, one year's labels were shuffled and the features rebuilt from scratch. Had anything been reading the answer the scores would have collapsed, and they moved by as much as chance predicts.

Splitting by time

Fire years are not drawn independently. A random draw would result in 2024 in the training set, while 2023 would be in the test set, thus allowing the model to peek into its future; consequently, the dataset is split chronologically: 2005-2019 for training, 2020-2022 for tuning, and 2023-2025 in the held-out dataset. The held-out data years were scored only once, at the end, after all feature selections and modelling decisions were made.

SECTION 3

Algorithms and model overview

The six learners were trained on the same row data and evaluated on the same left-out years. Four of them are the ones whose names appear in the assessment brief, and the other two were introduced as a challenge for the comparison.

All six models and both baselines were built in Python, using scikit-learn for the classical models (Pedregosa et al., 2011), Keras for the neural networks (Chollet, 2015), NumPy for the arrays (Harris et al., 2020) and Matplotlib for the charts (Hunter, 2007).

Decision tree

A single CART tree. The readable baseline: every prediction can be traced to a chain of thresholds.

Random forest

200 trees on bootstrapped samples, voting. Handles the interactions between rainfall, greenness and fire history without being told they exist.

Naive Bayes

Gaussian naive Bayes. Assumes the features are independent given the class, which they are not, so it sets a probabilistic floor.

Multilayer ANN

A dense feed-forward network trained with Keras (Chollet, 2015) on the same 50 features.

LSTM

A recurrent network reading each cell's 26-year sequence, with a second input branch for the static weather and greenness features (Hochreiter & Schmidhuber, 1997).

Gradient boosting

Histogram-based boosted trees, each correcting the last (Friedman, 2001). This is the model that won.

There are two baselines below all of the six. The first one follows what happened in that cell last year, and the second one always gives the answer 'not burnt'. An algorithm that fails to perform better than both of them has not learned anything worthwhile, especially the second one.

Correcting for imbalance

Without the weight, all models mostly gave up predictions for late fires since it is more economical to be correct for the general case. The use of class weights makes such a strategy very costly. With class weights used, the F1 measure for late fires increases from 0.420 to 0.553 in gradient boosting, on the same dataset and split, whereas overall accuracy shows little improvement. Both weighted and non-weighted versions of each model were trained and reported.

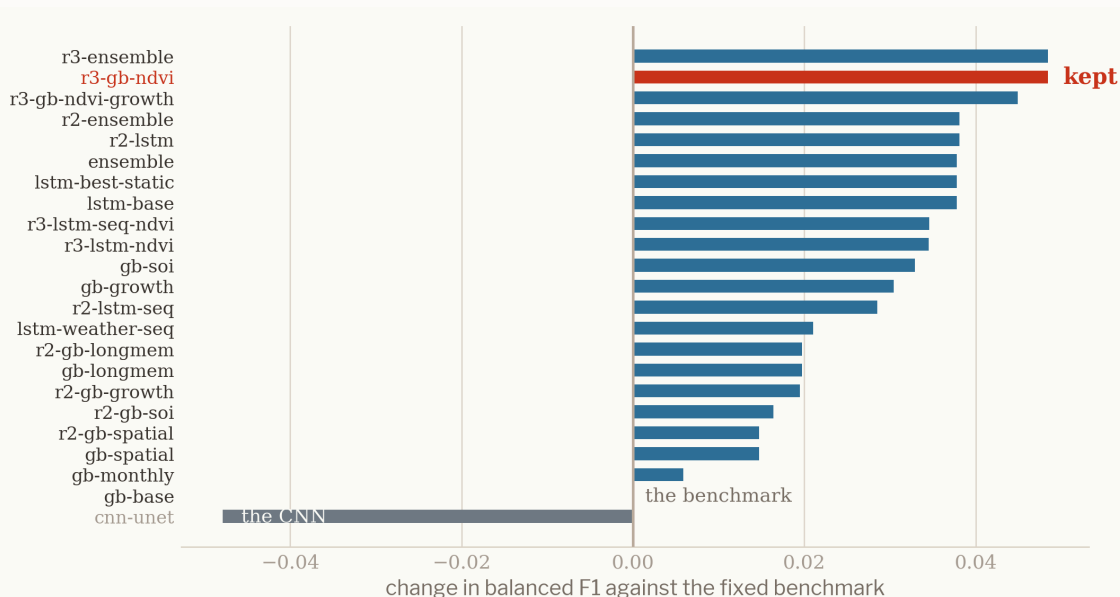
Systematic tuning

The first settings for each model were picked by hand, so every model was then tuned with a proper search. The decision tree, random forest and gradient boosting used a randomised search (Bergstra & Bengio, 2012), naive Bayes used a grid search, and the two neural networks used a small grid over their size, dropout and learning rate. Every candidate was trained on the training years and scored only on the validation years (2020–2022). The best setting was then refit on all the training years and tested once on 2023–2025.

Model	Search	Settings tried	Validation F1	Test F1 tuned	Test F1 by hand
Gradient boosting	Randomised	14	0.679	0.684	0.684
LSTM	Grid	4	0.673	0.670	0.670
Multilayer ANN	Grid	7	0.678	0.674	0.667
Random forest	Randomised	10	0.685	0.662	0.660
Decision tree	Randomised	25	0.643	0.632	0.604
Naive Bayes	Grid	11	0.639	0.603	0.602

Table 2. Systematic tuning. Settings were chosen on the validation years only, and the test years were used once. Neural network scores vary by a few thousandths between runs, so the LSTM scores 0.670 here against 0.673 in Table 1.

Tuning changed little. Gradient boosting scored 0.684 on the test years with both the hand-picked and the tuned settings, so it stays the best model. The decision tree gained the most, from 0.604 to 0.632, because the search chose much larger leaves (500 rows each), which stops it memorising the training years. No other model moved by more than 0.007. The only change in order is that the tuned multilayer ANN (0.674) edges past the LSTM (0.670).



Twenty-three experiments, each scored against the same fixed benchmark and recorded as kept or reverted. One was kept.

The final set of features was not chosen based on intuition. There were twenty-three trials conducted against a fixed benchmark and recorded in any case, even the U-Net model (Ronneberger et al., 2015) that interpreted fire information as an image instead of a tabular dataset. This model scored 0.588, which is 0.048 worse than the benchmark; hence, it was rolled back.

SECTION 4

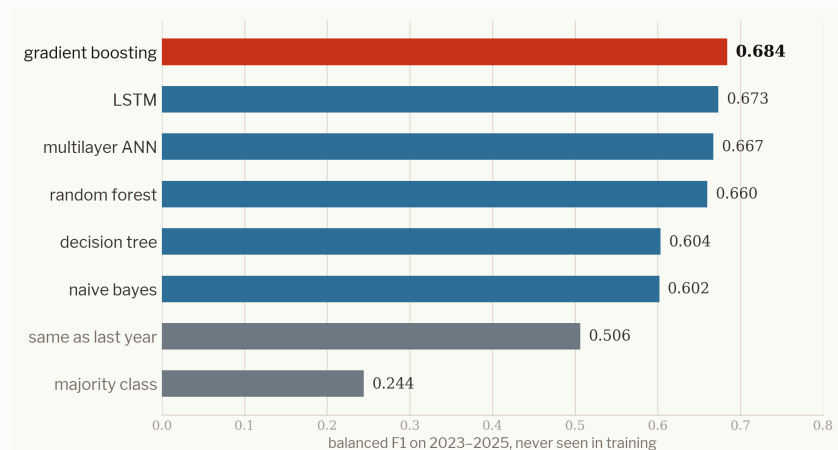
Model performance evaluation

All values in the figures below correspond to 2023, 2024, and 2025, unseen by any trained and tuned model. F1 score, precision, and recall are macro-averaged, meaning that each class carries the same weight and cannot be ignored by a model to perform well. Balanced versions of the models are provided because of better results.

Model	Accuracy	Precision	Recall	F1	F1 on late fires
Gradient boosting	0.725	0.679	0.690	0.684	0.553
LSTM	0.692	0.679	0.687	0.673	0.561
Multilayer ANN	0.678	0.670	0.695	0.667	0.556
Random forest	0.727	0.688	0.651	0.660	0.468
Decision tree	0.628	0.598	0.632	0.604	0.495
Naive Bayes	0.644	0.592	0.616	0.602	0.398
Baseline: same as last year	0.567	0.511	0.504	0.506	0.241
Baseline: always unburnt	0.578	0.193	0.333	0.244	0.000

Table 1. Held-out performance on 2023–2025. The final column is the class that matters most operationally.

The gradient boosting outperforms other models in terms of F1 score with 0.684 and accuracy equal to 0.725. The LSTM comes next with an F1 score of 0.673. The gap to read is not between the models but between the models and the naive rule, which is getting last year's result, resulting in an F1 score of 0.506 on exactly the same data, so gradient boosting adds 0.178 over what fire officers know without our help.



Balanced F1 on the held-out years. The two grey bars are the baselines.

Every model finds late fires hardest. Gradient boosting reaches 0.553 on that class against 0.816 on unburnt, and its confusion matrix shows why: of the 35,285 squares that truly burned after July, it calls 10,478 of them no fire. Late fires are the rarest class and the most driven by weather after April, which is exactly the information the model is not allowed to see.

Scoring the probabilities: ROC-AUC and PR-AUC

Accuracy, precision, recall and F1 only judge the final answer. Each model also gives a probability for each class, and ROC-AUC and PR-AUC judge those probabilities at every possible cut-off. ROC-AUC is the chance that the model ranks a square from a class above a square from outside it, so guessing scores 0.5. PR-AUC is stricter when a class is rare (Saito & Rehmsmeier, 2015): guessing scores the class's share of the data, which for late fires is 0.255, not 0.5.

Model	ROC-AUC	PR-AUC	PR-AUC early	PR-AUC late	Test F1
Gradient boosting	0.862	0.732	0.731	0.558	0.684
LSTM	0.854	0.724	0.735	0.534	0.670
Multilayer ANN	0.856	0.713	0.724	0.505	0.667
Random forest	0.859	0.730	0.731	0.551	0.660
Decision tree	0.793	0.633	0.646	0.413	0.604
Naive Bayes	0.797	0.637	0.676	0.390	0.602

Table 3. ROC-AUC and PR-AUC on 2023-2025, each class against the rest and averaged. Guessing would score 0.5 on ROC-AUC, and 0.166 (early) and 0.255 (late) on PR-AUC.

Gradient boosting ranks best on both, with a ROC-AUC of 0.862 and a PR-AUC of 0.732. The random forest is a close second on both (0.859 and 0.730) even though its F1 is lower, which means its probabilities rank squares well but its default cut-off calls too few late fires. On late fires gradient boosting reaches a PR-AUC of 0.558, a little over twice what guessing would score. That is useful, and it confirms what the F1 scores already show: late fires remain the hardest class.

Checking for overfitting

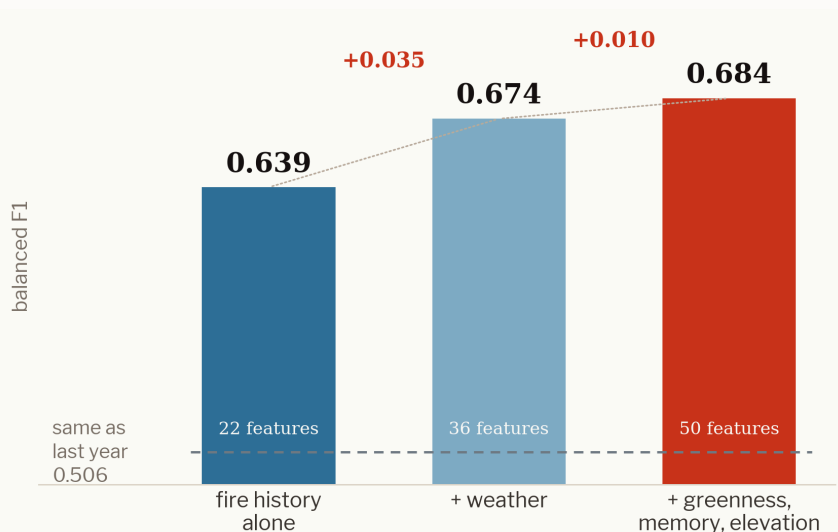
A model that has memorised its training data scores far higher on the years it learned from than on years it has not seen. Table 4 compares the two for every model.

Model	Train F1	Validation F1	Test F1	Train minus test	Stopped at epoch
Gradient boosting	0.821	0.678	0.684	0.137	—
LSTM	0.687	0.671	0.670	0.016	3
Multilayer ANN	0.702	0.674	0.667	0.035	11
Random forest	0.924	0.685	0.660	0.264	—
Decision tree	0.700	0.627	0.604	0.096	—
Naive Bayes	0.581	0.641	0.602	-0.021	—

Table 4. Macro F1 on the training years (2005–2019), the validation years (2020–2022) and the test years (2023–2025).

The random forest shows the clearest overfitting: 0.924 on the training years against 0.660 on the test years. Its trees keep splitting until a leaf holds as few as five rows, so they learn much of the training years by heart. Gradient boosting has a smaller gap, 0.821 against 0.684. The two neural networks have the smallest gaps because training stops at the epoch with the best validation score, epoch 11 for the ANN and epoch 3 for the LSTM, before they learn the training years too closely. Naive Bayes scores lower on its training years than on the test years, the sign of a model too simple to overfit. For every model the validation and test scores sit within 0.04 of each other, so the scores hold up on a second set of unseen years and not only on the years used to choose settings.

What each source of data bought

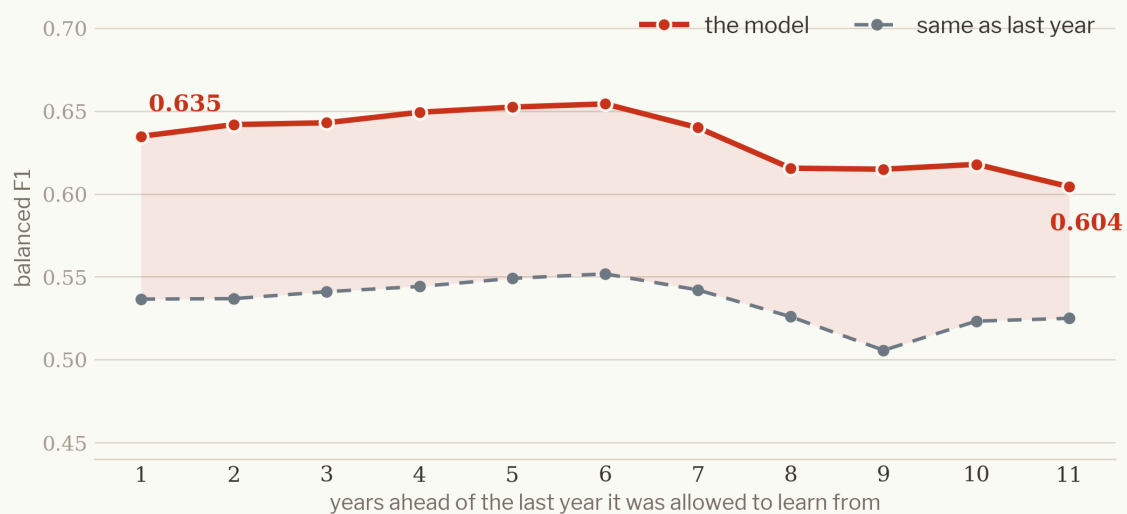


Fire history alone, then weather, then greenness with longer memory and elevation.

Fire history on its own reaches 0.639. Adding weather takes it to 0.674, the single largest step in the project. The third run adds greenness together with the longer memory features and elevation for a further 0.010; isolated in the experiment ledger, greenness on its own is worth 0.029. It was kept because the gain holds on years the model never saw and because two of the model's five strongest inputs turn out to be greenness measures, though the Earth Engine extraction cost about ten minutes a run against sixty seconds for everything else.

Testing it sixty-six times

One good test year could be luck, so the record was cut at each year from 2014 to 2024, the model was given only what came before the cut, and it was scored on every year after it. That is 66 separate tests.



Skill against how far ahead the model was asked to guess, averaged over 66 folds.

Skill does not fade with distance. Guessing one year ahead scores 0.635; guessing eleven years ahead it still scores 0.604, with the naive baseline well below it throughout. That is not the model being clever. It says the strongest clue is the wet season just gone, which is known by the end of April however old the fire history behind it is.

Across the 66 tests the model averaged 0.640, with a standard deviation of 0.033 and a range of 0.577 to 0.689, and it beat the same-as-last-year baseline in 66 of the 66.

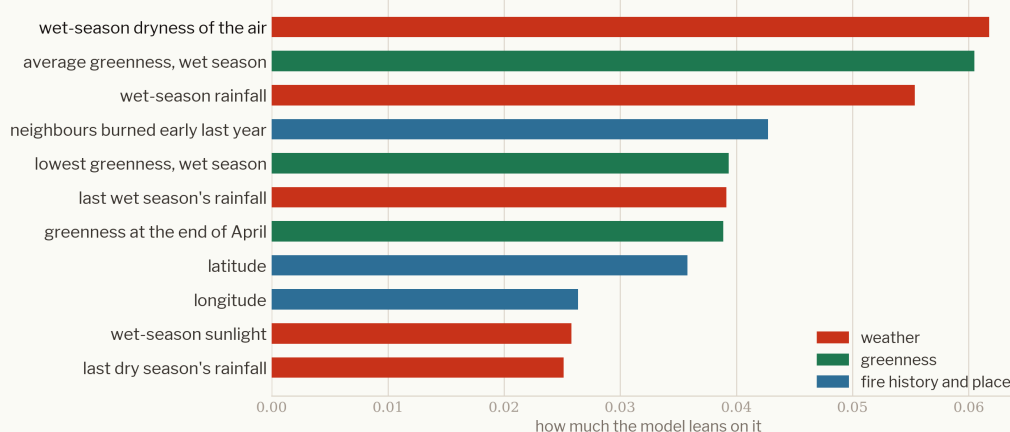
Why not ordinary k-fold cross-validation?

Cross-validation splits the data into k parts, trains on all but one, tests on the part left out, and repeats until every part has been tested once. Repeated stratified k-fold does this several times with random splits that keep the class mix the same in every part. That works when rows are independent. Here they are not: a square in 2012 looks like its neighbour in 2012 and like itself in 2013, so a random split puts near-copies of the test rows into training (Roberts et al., 2017).

To measure how much this matters, gradient boosting was run through repeated stratified 5-fold cross-validation, repeated twice for 10 folds in all, on a 240,000-row sample of the training years. It scored 0.772 ± 0.004 . The same model trained on the same rows scored 0.679 on the validation years and

0.677 on the test years. Random cross-validation would have overstated the score by about 0.09, which is why every result in this report comes from years the model had not seen, and why the walk-forward tests above take the place of ordinary cross-validation.

What the model leans on



Random forest impurity importance, coloured by where each number came from.

Nobody told the model that a wet, hot wet season grows the grass that carries an October fire. Its strongest inputs are the dryness of the air through the wet, average greenness, wet-season rainfall and what the neighbouring country did last year. That it lands close to where fire ecology already sits suggests that it learned the problem rather than the dataset.

Impurity importance comes from the forest's own training and can favour inputs with many distinct values (Strobl et al., 2007), so it was checked with permutation importance (Breiman, 2001). Each input was shuffled in turn on 40,000 held-out squares, and the fall in macro F1 was recorded, five times over. For the random forest, 7 of its 10 highest-ranked inputs are the same under both methods, led by wet-season rainfall, the dryness of the air through the wet, average wet-season greenness. Gradient boosting leans most on average wet-season greenness, April greenness, latitude, wet-season rainfall, and 8 of its top 10 appear in the forest's chart. Both methods and both models point to the same thing: the wet season just gone.

SECTION 5

Limitations

Late fires

At 0.553 the class that matters most is still the weakest. It is the rarest and the most weather-driven, and weather after April is what the 30 April rule deliberately excludes.

A season, not a day

The model says which squares are likely to burn and roughly when. It is not a fire-danger rating and should not be read as one.

2026 is unverified

No burnt-area record exists for a year still running, so the forecast is published as a claim to be checked in 2027 rather than as a result.

One Territory, one grid

Everything is fitted to the Northern Territory at 0.05 degrees. Whether it transfers to the Kimberley or Cape York is an open question, not an assumption.

Chosen on the same years

All the tuning and the 23 experiments were judged on the same validation years, 2020–2022, so some of the gain may fit those three years rather than fire in general. The test years were used only once, and the walk-forward tests guard against a lucky split, but later years are the real check.

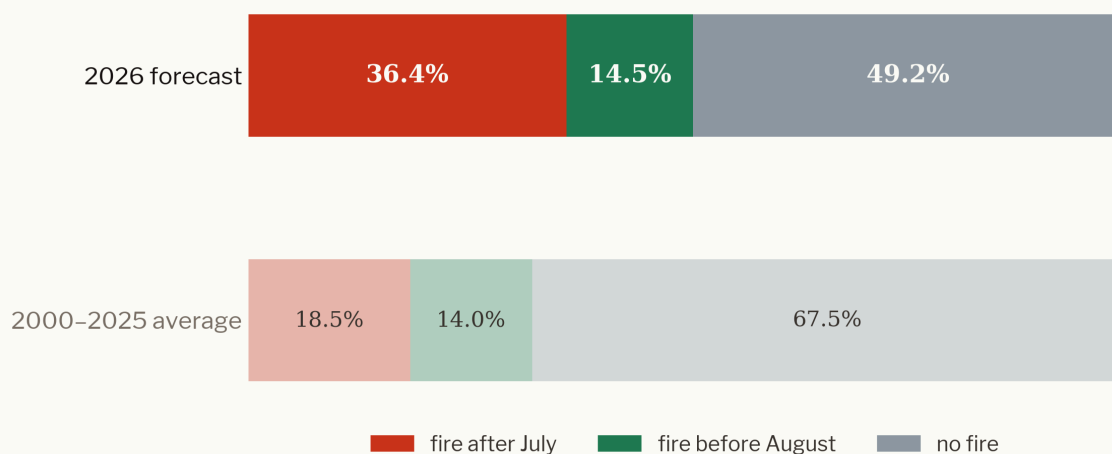
No outside check

Every score here is measured against NAFI's own fire scars. The model has not been compared with an independent record, such as ranger burn logs or another agency's fire mapping, so any mistakes in NAFI's mapping are learned as if they were true.

SECTION 6

Conclusion

The gradient boosting algorithm identified three regimes of fire at macro F1 of 0.684 and accuracy of 0.725 on three years it had never seen before, which is 0.178 higher than the naive strategy of predicting that next year will be the same as the previous year. Results from 66 walk-forward tests show that this is not a result of one lucky split. Weather proved to be the most important source of data, with greenness providing some additional value.



The 2026 forecast against the 26-year average.

Applied to 2026, the model expects 51% of the Territory to burn against a 26-year average of 33%, with 15% before August and 36% after July, at a mean confidence of 0.733. It reads that from a wet season of 783 mm, 45% above the 2005 to 2019 average and the second wettest in 22 years, and the greenest April in the record. More water grew more grass, and more grass is more fuel.

The model and its evidence are published as a public site so the forecast can be read by someone who is not a data scientist, and every sentence on that site is computed from the model output rather than typed in. The forecast can be checked against the satellite record in 2027.

Live site

pyrantis.harshlabs.workers.dev

Source code

github.com/harshrastogii/PYRANTIS

SECTION 7

Individual contributions

Harsh Rastogi (386401)

Project lead. Problem framing and the choice of the three-class target, the data pipeline across all three sources, the 2026 forecast, and the public web product. Presents slides 1–5 and 19–22.

Saira Zafar (407193)

Labelling from the NAFI rasters, feature engineering and the 30 April rule, the two leakage tests and the temporal split design. Presents slides 6–11.

Tharushi Wimalachandra (387594)

Model training and tuning across all six classifiers, the experiment ledger, walk-forward validation and the results analysis. Presents slides 12–18.

All three members reviewed each other's work, and the presentation runs to about 14 minutes with no member exceeding six.

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